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Systems and methods for condition monitoring of top drive internal blowout preventer valves

Abstract

A system includes a top drive including a handling ring assembly including a main body, a main shaft that traverses the main body of the handling ring assembly, an internal blowout preventer (IBOP) valve, wherein the main shaft engages the IBOP valve, a two-piece actuator sleeve assembly including a non-rotating portion and a rotating portion, at least one hydraulic cylinder connected to the main body of the handling ring assembly, wherein the at least one hydraulic cylinder actuates the two-piece actuator sleeve assembly, and left and right hand crank assemblies having left and right wireless encoders connected thereto, wherein the crank assemblies with the wireless encoders are attached to, and are configured to rotate with, the IBOP valve along with the rotating portion of the two-piece actuator sleeve assembly.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) The present document is a National Stage Entry of International Application No. PCT/US2022/048826, filed Nov. 3, 2022, which is based on and claims priority to U.S. Provisional Patent Application No. 63/276,585, filed Nov. 6, 2021, which is incorporated herein by reference in its entirety.

BACKGROUND

(1) Top drives, which are drilling machines suspended from a block and tackle arrangement in a mast/derrick of a drilling rig, perform many functions related to rotary drilling. One of the primary functions of the top drive is to serve as a conduit for drilling fluid, which enters the top drive through a rotary drilling hose from a high-pressure mud standpipe, which then passes through a rotary seal called a washpipe, then into a bore of a main shaft of the top drive, and then down into the wellbore through the drilling tubular.

(2) A valve called an IBOP (“internal blowout preventer”) is placed in line between the top drive main shaft and the drilling tubular. The IBOP is a high pressure ball valve that serves two primary

functions: (1) closing off the path for drilling fluid during the connection of new tubulars (called a “mud saver” in this case), and (2) closing in the event of a “kick” (a type of loss of well control, where high pressure gas goes back up the drill string).

(3) IBOP valves often have reduced lives because actuator designs are unable to ensure that the valves are fully open or fully closed for different reasons (i.e., lack of adjustment, incorrect travel, etc.). Replacing an IBOP valve is a time consuming and difficult task, and new valves are expensive. Accordingly, there is a need to improve the life of IBOP valves and track valve degradation so that total valve failure can be anticipated and planned for, rather than merely reacted to.

SUMMARY

(4) A system according to one or more embodiments of the present disclosure includes, a top drive including a handling ring assembly including a main body, a main shaft that traverses the main body of the handling ring assembly, an internal blowout preventer valve, wherein the main shaft engages the internal blowout preventer valve, a two-piece actuator sleeve assembly including a non-rotating portion and a rotating portion, at least one hydraulic cylinder connected to the main body of the handling ring assembly, wherein the at least one hydraulic cylinder actuates the two-piece actuator sleeve assembly, a left hand crank assembly having a left wireless encoder connected thereto, and a right hand crank assembly having a right wireless encoder connected thereto, wherein the crank assemblies with the wireless encoders are attached to, and are configured to rotate with, the internal blowout preventer valve along with the rotating portion of the two-piece actuator sleeve assembly.

(5) A system for condition monitoring an internal blowout preventer valve of a top drive, the system including a crank assembly that interfaces with a ball valve inside a body of the internal blowout preventer valve, a wireless encoder connected to the crank assembly via a crank/encoder interface, a processing device communicatively connected with the wireless encoder, and a human machine interface that visually displays position measurement data calculated by the processing device.

(6) However, many modifications are possible without materially departing from the teachings of this disclosure. Accordingly, such modifications are intended to be included within the scope of this disclosure as defined in the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Certain embodiments of the disclosure will hereafter be described with reference to the accompanying drawings, wherein like reference numerals denote like elements. It should be understood, however, that the accompanying figures illustrate the various implementations described herein and are not meant to limit the scope of various technologies described herein, and:

(2) FIG. 1 generally depicts a well construction system including a drilling system having a top drive according to one or more embodiments of the present disclosure;

(3) FIG. 2 is a block diagram of various components of a top drive according to one or more embodiments of the present disclosure;

(4) FIGS. 3A and 3B are front views of certain components of a top drive in retracted and extended positions including an IBOP valve with some components covered in operation, according to one or more embodiments of the present disclosure;

(5) FIGS. 4A and 4B are front views of certain components of a top drive in retracted and extended positions including an IBOP valve with transparent covers revealing the components covered in FIGS. 3A and 3B, according to one or more embodiments of the present disclosure;

(6) FIGS. 5A and 5B are side views of certain components of a top drive in retracted and extended

positions including an IBOP valve with transparent covers, according to one or more embodiments of the present disclosure;

(7) FIGS. **6A** and **6B** are isometric views of certain components of a top drive in retracted and extended positions including an IBOP valve, according to one or more embodiments of the present disclosure;

(8) FIGS. **7A** and **7B** are additional isometric views of certain components of a top drive in retracted and extended positions including an IBOP valve, according to one or more embodiments of the present disclosure;

(9) FIGS. **8A** and **8B** are additional isometric views of certain components of a top drive in retracted and extended positions including an IBOP valve, according to one or more embodiments of the present disclosure;

(10) FIGS. **9A** and **9B** are section views of certain components of a top drive in retracted and extended positions including an IBOP valve, according to one or more embodiments of the present disclosure;

(11) FIGS. **10A** and **10B** are zoomed in section views of certain components of a top drive in retracted and extended positions including an IBOP valve, according to one or more embodiments of the present disclosure;

(12) FIG. **11** shows a detailed, zoomed in section view of certain components of a top drive in a retracted position including an IBOP valve, according to one or more embodiments of the present disclosure;

(13) FIGS. **12A-12C** show a general arrangement of the IBOP valve, according to one or more embodiments of the present disclosure;

(14) FIGS. **13A** and **13B** show zoomed in section views of movement of certain components of a top drive in retracted and extended positions, according to one or more embodiments of the present disclosure;

(15) FIG. **14A** shows a perspective view of a crank assembly rotated in a first direction, according to one or more embodiments of the present disclosure;

(16) FIG. **14B** shows a partial cutaway view of the crank assembly shown in FIG. **14A**, according to one or more embodiments of the present disclosure;

(17) FIG. **14C** shows a perspective view of a crank assembly rotated in a second direction, according to one or more embodiments of the present disclosure;

(18) FIGS. **14D** and **14E** show bottom perspective views of a crank assembly rotated in the first direction and the second direction, respectively, according to one or more embodiments of the present disclosure;

(19) FIGS. **15A** and **15B** show front and perspective views of an encoder assembly within a guard, according to one or more embodiments of the present disclosure;

(20) FIG. **15C** shows a detailed perspective view of an encoder assembly within a transparent guard, according to one or more embodiments of the present disclosure;

(21) FIG. **15D** shows a perspective view of an anti-rotation bracket with rollers of an encoder assembly, according to one or more embodiments of the present disclosure;

(22) FIGS. **16A** and **16B** show zoomed in partial views of certain components of a top drive including a crank valve in open and closed positions, according to one or more embodiments of the present disclosure;

(23) FIGS. **17A** and **17B** show zoomed in partial, angled cutaway views of certain components of a top drive including a bearing carrier roller feature, according to one or more embodiments of the present disclosure;

(24) FIGS. **18A** and **18B** show zoomed in partial views of certain components of a top drive including a crank/encoder guard, according to one or more embodiments of the present disclosure;

(25) FIGS. **19A** and **19B** show zoomed in partial views of certain components of the top drive as shown in FIGS. **18A** and **18B** with the links removed and the encoder guards transparent for better

visibility, according to one or more embodiments of the present disclosure;

(26) FIGS. 20A and 20B show human machine interface (HMI) images of IBOP fully closed and fully open positions, according to one or more embodiments of the present disclosure;

(27) FIGS. 21A and 21B show HMI images of IBOP not fully closed and not fully open positions, according to one or more embodiments of the present disclosure.

(28) FIGS. 22A and 22B show HMI images of IBOP not fully closed and not fully open positions along with a depiction of reduction of the operating extent of the IBOP valve over time, according to one or more embodiments of the present disclosure; and

(29) FIGS. 23A and 23B show HMI images of IBOP not fully closed and not fully open positions along with a depiction of reduction of the operating extent of the IBOP valve such that replacement is required, according to one or more embodiments of the present disclosure.

DETAILED DESCRIPTION

(30) In the following description, numerous details are set forth to provide an understanding of some embodiments of the present disclosure. However, it will be understood by those of ordinary skill in the art that the system and/or methodology may be practiced without these details and that numerous variations or modifications from the described embodiments may be possible.

(31) In the specification and appended claims, the terms “connect,” “connection,” “connected,” “in connection with,” and “connecting,” are used to mean “in direct connection with,” in connection with via one or more elements.” The terms “couple,” “coupled,” “coupled with,” “coupled together,” and “coupling” are used to mean “directly coupled together,” or “coupled together via one or more elements.” The term “set” is used to mean setting “one element” or “more than one element.” As used herein, the terms “up” and “down,” “upper” and “lower,” “upwardly” and “downwardly,” “upstream” and “downstream,” “uphole” and “downhole,” “above” and “below,” “top” and “bottom,” and other like terms indicating relative positions above or below a given point or element are used in this description to more clearly describe some embodiments of the disclosure. Commonly, these terms relate to a reference point at the surface from which drilling operations are initiated as being the top point and the total depth being the lowest point, wherein the well (e.g., wellbore, borehole) is vertical, horizontal, or slanted relative to the surface.

(32) The present disclosure generally relates to top drive IBOP valves. More specifically, one or more embodiments of the present disclosure are related to condition monitoring of top drive IBOP valves. Condition monitoring is a process of monitoring equipment condition indicators for changes to identify future faults, failures, breakdowns, and other maintenance problems associated with equipment. Condition monitoring is increasingly utilized in the oil and gas industry as part of predictive maintenance of wellsite (e.g., drilling) equipment. Condition monitoring utilizes condition data generated by peripheral (e.g., add-on) sensors and instruments to gain more insight to future maintenance problems. Condition data, such as vibration data, acoustic data, thermographic (e.g., infrared signature) data, is used solely to indicate condition of equipment. Condition monitoring also includes analyzing operational data to determine amount of equipment usage and compare the determined equipment usage to expected operational lifetime specifications and/or calculations. According to one or more embodiments of the present disclosure, condition monitoring may be used determine whether a top drive IBOP valve is fully open or fully closed so as to not wash out the valve during drilling operations. Condition monitoring may also be used to track degradation of the valve, for example. With respect to condition monitoring, this disclosure is related to U.S. Patent Application Publication No. 2020/0291767, entitled “PERFORMANCE BASED CONDITION MONITORING,” the disclosure of which is incorporated herein by reference in its entirety.

(33) Most top drives are equipped with two IBOP valves for redundancy—one remotely operated, and one manually operated. These IBOP valves are typically metal-to-metal seal ball valves with a 90° range of motion. At one extent, the IBOP valve will be fully closed, and at the other, fully opened. Opening and closing fully is critical to long term life of the IBOP valve because of the type

of fluid, the flow rates, and pressures associated with the application.

(34) One or more embodiments of the present disclosure includes an IBOP actuator design that is simple to adjust and to operate. According to one or more embodiments of the present disclosure, the IBOP actuator design is equipped with wireless incremental encoders installed on the cranks of the valve, so that instantaneous valve position is always known, and can be tracked over time as the operating extent of the valve is progressively reduced by wear, damage, and other application specific factors.

(35) Referring now to FIG. 1, a well construction system **100** including a drilling system **10** is shown according to one or more embodiments of the present disclosure. Notably, the system **10** may be operated to drill a well **12** to access a subterranean resource, such as oil or natural gas. As depicted, the system **10** includes an onshore drilling rig **14**, although the system **10** could instead be an offshore system in other embodiments. The drilling rig **14** uses a drill string **16** and a drill bit **18** to form the well **12**. It will be appreciated that the drill string **16** can include various members, such as drill pipes, tool joints, drill collars, and a saver sub that prevents wear on a threaded connection of a rotating system (e.g., a top drive) that drives rotation of the drill string **16**.

(36) The drilling rig **14** also includes a mast **20** and a hoisting system (here generally shown as including a traveling block **22**, a crown block **24**, and draw works **26**) to enable a top drive **28** to be raised and lowered with respect to a drill floor **30**. The drill string **16** is suspended from the top drive **28** through a hole in the drill floor **30** and through surface equipment (e.g., a blowout preventer **32** in the cellar). The drill string **16** can be rotated by the top drive **28** and can be raised and lowered with the top drive **28** (via the traveling block **22**) to facilitate drilling operations.

(37) Still referring to FIG. 1, the well construction system **100** may also include a control center **190** from which the top drive **28** may be monitored and controlled. The control center **190** may be located on the drill floor **30** or another location of the well construction system **100**, such as a surface of the wellsite, for example. The control center **190** may include a facility **191** (e.g., a room, a cabin, a trailer, etc.) containing a control workstation **197**, which may be operated by a human wellsite operator **195** to monitor and control various wellsite equipment or portions of the well construction system **100**. The control workstation **197** may include or be communicatively connected with a processing device **192** (e.g., a controller, a computer, etc.), such as may be operable to receive, process, and output information to monitor operations of and provide control to one or more portions of the well construction system **100**. For example, the processing device **192** according to one or more embodiments of the present disclosure may be communicatively connected with two wireless encoders installed in two crank assemblies of the top drive **28** as described herein, and may be operable to receive signals from and transmit signals to such equipment to perform various operation described herein. The processing device **192** may store executable program code, instructions, and/or operational parameters or set-points, including for implementing one or more aspects of methods and operations described herein. The processing device **192** may be located within and/or outside of the facility **191**.

(38) The control workstation **197** may be operable for entering or otherwise communicating control commands to the processing device **192** by the wellsite operator **195**, and for displaying or otherwise communicating information from the processing device **192** to the wellsite operator **195**. The control workstation **197** may include a plurality of human-machine interface (HMI) devices, including one or more input devices **194** (e.g., a keyboard, a mouse, a joystick, a touchscreen, etc.) and one or more output devices **196** (e.g., a video monitor, a touchscreen, a printer, audio speakers, etc.). Communication between the processing device **192**, the input and output devices **194**, **196**, and the various wellsite equipment may be via wired and/or wireless communication means.

(39) One example of a top drive **28** is generally depicted in FIG. 2. In this embodiment, the top drive **28** includes a connector **40** for attaching the top drive **28** to the traveling block **22**. A drive stem **46** is suspended from a swivel **42** through a motor **44**, which drives rotation of the drive stem **46** within the top drive **28**. The drive stem **46** (which is sometimes referred to as a main shaft or a

quill) can be connected to a drill string **16** to cause the drill string **16** to rotate along with the drive stem **46**. The top drive **28** of FIG. 2 also includes a handling ring **48** connected to a pipe handler **50** and to an elevator **52**.

(40) Referring now to FIGS. 3A and 3B, front views of certain components of a top drive **28** are shown, according to one or more embodiments of the present disclosure. Specifically, FIG. 3A shows the top drive **28** in a retracted position, and FIG. 3B shows the top drive **28** in an extended position, as further described below. As shown in FIGS. 3A and 3B, the top drive **28** includes a motor **44**, as previously described, a handling ring assembly **48** including a main body **49**, and a drive stem (or main shaft) **46** that traverses the main body **49** of the handling ring assembly **48**. As shown in FIGS. 3A and 3B, in addition to the main body **49**, the handling ring assembly **48** also includes arms **51**. As further shown in FIGS. 3A and 3B, the top drive **28** also includes an internal blowout preventer (IBOP) valve **53**, and the main shaft **46** of the top drive **28** engages with the IBOP valve **53**, as further described below.

(41) As also shown in FIGS. 3A and 3B, the top drive **28** also includes a pair of hydraulic cylinders **56**, a two-piece actuator sleeve assembly **54**, and a pair of guards or covers **62a**, **62b**, which conceal crank assemblies and connected wireless encoders, as further described below. As also shown in FIGS. 3A and 3B, the top drive **28** may include a plurality of tool joint safety clamps **64** disposed along the IBOP valve **53**, for example. Moreover, the top drive **28** may include links **66** attached to the arms **51** of the handling ring assembly **48**, and an elevator **52** connected to the links **66**.

(42) FIGS. 4A and 4B show front views of the same components of the top drive **28** shown in FIGS. 3A and 3B, for example. FIG. 4A shows the top drive **28** in a retracted position, and FIG. 4B shows the top drive **28** in an extended position, as further described below. In FIGS. 4A and 4B, however, the guards or covers **62a** and **62b** shown in FIGS. 3A and 3B are transparent, such that the left and right wireless encoders **60a**, **60b** are visible. As further described below, the left and right wireless encoders **60a**, **60b** are connected to corresponding left hand and right hand crank assemblies. FIGS. 4A and 4B also more clearly show the two pieces of the two-piece actuator sleeve assembly **54**, which includes a non-rotating portion **54a** and a rotating portion **54b**, the operation of which are further described below.

(43) To provide additional views to facilitate understanding of one or more embodiments of the present disclosure, FIGS. 5A and 5B show side views of certain components of the top drive **28**, as previously described in view of FIGS. 3A, 3B, 4A, and 4B, for example. Specifically, FIG. 5A shows the top drive **28** in a retracted position, and FIG. 5B shows the top drive **28** in an extended position, as further described below. Similar to FIGS. 4A and 4B, the side views of the top drive **28** shown in FIGS. 5A and 5B show transparent guards or covers **62a** and **62b** to enhance visibility of the non-rotating portion **54a** and the rotating portion **54b** of the two-piece actuator sleeve assembly **54**, for example. Even with transparent guards or covers **62a**, **62b**, however, in the side views shown in FIGS. 5A and 5B, the links **66** of the top drive **28** obstruct the view of the left and right wireless encoders **60a**, **60b**, which are connected to corresponding left hand and right hand crank assemblies, as further described below. With respect to FIGS. 5A and 5B, like components of the top drive **28** are labeled with like reference numerals, as previously described.

(44) To provide additional views to facilitate understanding of one or more embodiments of the present disclosure, FIGS. 6A, 6B, 7A, 7B, 8A, and 8B show different isometric views of certain components of the top drive **28**, as previously described in view of FIGS. 3A, 3B, 4A, and 4B for example. Specifically, FIGS. 6A and 6B show an isometric view of the top drive **28** in a retracted position and an extended position, respectively. FIGS. 7A and 7B show a different isometric view of the top drive **28** in a retracted position and an extended position, respectively. Finally, FIGS. 8A and 8B show a still different isometric view of the top drive **28** in a retracted position and an extended position, respectively. With respect to FIGS. 6A, 6B, 7A, 7B, 8A, and 8B, like components of the top drive **28** are labeled with like reference numerals, as previously described.

(45) For additional perspective, FIGS. **9A** and **9B** provide partial section views of certain components of a top drive **28** according to one or more embodiments of the present disclosure. Specifically, FIG. **9A** shows the top drive **28** in a retracted position, and FIG. **9B** shows the top drive **28** in an extended position, as further described below. With respect to FIGS. **9A** and **9B**, like components of the top drive **28** are labeled with like reference numerals, as previously described. Moreover, FIGS. **9A** and **9B** also show that the top drive **28** includes left hand and right hand crank assemblies **58a**, **58b** having the left and right wireless encoders **60a**, **60b** connected thereto. According to one or more embodiments of the present disclosure, the crank assemblies **58a**, **58b** with the wireless encoders **60a**, **60b** are attached to, and are configured to rotate with, the internal blowout preventer valve **53** along with the rotating portion **54b** of the two-piece actuator sleeve assembly **54**, as further described below.

(46) FIGS. **10A** and **10B** provide zoomed in views of the partial section views of certain components of the top drive **28** shown in FIGS. **9A** and **9B**, according to one or more embodiments of the present disclosure. In this way, FIGS. **10A** and **10B** provide additional detail regarding the certain components of the top drive **28**, as previously described. Specifically, FIG. **10A** shows the top drive **28** in a retracted position, and FIG. **10B** shows the top drive **28** in an extended position, as further described below. With respect to FIGS. **10A** and **10B**, like components of the top drive **28** are labeled with like reference numerals, as previously described.

(47) FIG. **11** provides a further zoomed in view of certain components of the top drive **28** with the top drive **28** in the retracted position, as shown in FIG. **10A**. With respect to FIG. **11**, like components of the top drive **28** are labeled with like reference numerals, as previously described. As further shown, FIG. **11** shows a detailed view of a bearing **55** disposed between the non-rotating portion **54a** and the rotating portion **54b** of the two-piece actuator sleeve assembly **54**, according to one or more embodiments of the present disclosure. Due to this configuration, the bearing allows relative motion with respect to certain components of the top drive **28** since the hydraulic cylinders **56** are unable to rotate.

(48) Further, FIG. **11** more clearly shows the main shaft **46** of the top drive **28** engaged with the IBOP valve **53**. Referring now to FIGS. **12A** to **12C**, a general arrangement of the IBOP valve **53** according to one or more embodiments of the present disclosure is shown. Specifically, FIG. **12A** shows a section view of the IBOP valve **53**, which includes a valve body **53a**, a box connection end **53b**, a pin connection end **53c**, a ball valve **68**, and a ball valve crank **70**. FIG. **12B** shows the IBOP valve **53** of FIG. **12A** along line **12B-12B**, and FIG. **12C** shows the IBOP actuator crank interface **72**, according to one or more embodiments of the present disclosure. As shown in FIG. **11**, the main shaft **46** of the top drive **28** engages the box connection end **53b** of the IBOP valve **53**. In view of FIGS. **11**, **12A**, **12B**, and **12C**, crank assemblies **58a**, **58b** may fit into the ball valve cranks **70** of the IBOP valve **53** via the IBOP actuator crank interfaces **72**. In this way, the crank assemblies **58a**, **58b** of the top drive **28** are able to interface with the ball valve **68** inside the body **53a** of the IBOP valve **53**, according to one or more embodiments of the present disclosure. As also shown in FIG. **11**, the left hand crank assembly **58a** has a left wireless encoder **60a** connected thereto, and the right hand crank assembly **58b** has a right wireless encoder **60b** connected thereto.

(49) Referring now to FIGS. **13A** and **13B**, zoomed in section views of axial and rotational movement of certain components of the top drive **28** are shown, according to one or more embodiments of the present disclosure. In FIGS. **13A** and **13B**, the links **66** of the top drive **28** are removed for the sake of clarity. According to one or more embodiments of the present disclosure, the two-piece actuator sleeve assembly **54** includes a non-rotating portion **54a** and a rotating portion **54b**, as previously described. Due to this configuration, when the main shaft **46** of the top drive **28** rotates, the IBOP valve **53** rotates, and the rotating portion **54b** of the two-piece actuator sleeve assembly **54** also rotates, as shown in FIGS. **13A** and **13B**. As further shown in FIGS. **13A** and **13B**, the top drive **28** also includes hydraulic cylinders **56** connected to the main body **49** of the handling ring assembly **48** and connected to the two-piece actuator sleeve assembly **54**, according

to one or more embodiments of the present disclosure. By retracting and extending, the hydraulic cylinders **56** actuate the two-piece actuator sleeve assembly **54**, which allows the top drive **28** to assume the different retracted and extended positions shown in FIGS. **13A** and **13B**, respectively. For example, the hydraulic cylinders **56** may be connected to a hydraulic system of the top drive **28**, which feeds into a distribution system for supplying hydraulic fluid to the hydraulic cylinders **56**. In this way, applying hydraulic pressure near the top of the hydraulic cylinders **56** causes the hydraulic cylinders **56** to retract, which causes the two-piece actuator sleeve assembly **54** connected to the hydraulic cylinders **56** to move axially upward along a longitudinal axis of the IBOP valve **53**, which results in the top drive **28** assuming the retracted position shown in FIG. **13A**. Further, applying hydraulic pressure near the bottom of the hydraulic cylinders **56** causes the hydraulic cylinders **56** to extend, which causes the two-piece actuator sleeve assembly **54** connected to the hydraulic cylinders **56** to move axially downward along the longitudinal axis of the IBOP valve **53**, which results in the top drive **28** assuming the extended position shown in FIG. **13B**. This axial movement of the two-piece actuator sleeve assembly **54** either up or down along the longitudinal axis of the IBOP valve **53** may occur as the main shaft **46** and the IBOP valve **53** of the top drive **28** rotate, for example. Specifically, this axial movement of the two-piece actuator sleeve assembly **54** includes the non-rotating portion **54a** moving axially along the longitudinal axis of the IBOP valve **53**, and the rotating portion **54b** moving axially along the longitudinal axis of the IBOP valve **53**, along with the non-rotating portion **54a**, as the rotating portion **54b** continues to rotate.

(50) Still referring to FIGS. **13A** and **13B**, the crank assemblies **58a**, **58b**, which interface with the ball valve **68** inside the body **53a** of the IBOP valve **53**, and which are attached to the rotating portion **54b** of the two-piece actuator sleeve assembly **54**, rotate to shift a position of the ball valve **68** inside the body **53a** of the IBOP valve **53**. For example, when the hydraulic cylinders **56** are retracted, as shown in FIG. **13A**, the crank assemblies **58a**, **58b** may be rotated in a direction so as to shift the position of the ball valve **68** inside the IBOP valve **53** to a closed position. Further, when the hydraulic cylinders **56** are extended, as shown in FIG. **13B**, the crank assemblies **58a**, **58b** may be rotated 90° in a second direction so as to shift the position of the ball valve **68** inside the IBOP valve **53** from the closed position to an open position.

(51) Referring now to FIGS. **14A** to **14E**, different views of a left crank assembly **58a** according to one or more embodiments of the present disclosure, as shown in the top drive **28** of FIGS. **13A** and **13B**, for example, are shown. Specifically, FIG. **14A** shows a perspective view of the left crank assembly **58a** rotated in a first direction, which may be a closed position, according to one or more embodiments of the present disclosure, for example. Further, FIG. **14B** shows a partial cutaway view of the left crank assembly **58a** shown in FIG. **14A**, according to one or more embodiments of the present disclosure. Specifically, FIG. **14B** shows that the left crank assembly **58a** may include a crank body **57a**, a crank handle **57b**, a crank shaft **57c**, a cam follower **57d**, at least one O-ring **57e**, a retaining ring **57f**, and a needle bearing **57g**, according to one or more embodiments of the present disclosure. Although not shown, the corresponding right crank assembly **58b** may include similar components, according to one or more embodiments of the present disclosure. FIG. **14C** shows a perspective view of the left crank assembly **58a** rotated in a second direction, which may be an open position, for example, according to one or more embodiments of the present disclosure. Finally, FIGS. **14D** and **14E** show bottom perspective views of the left crank assembly **58a** rotated in the first direction and the second direction, respectively, according to one or more embodiments of the present disclosure.

(52) Referring back to FIGS. **13A** and **13B**, a left wireless encoder **60a** is connected to the left hand crank assembly **58a** via a left crank/encoder interface **59a**, and a right wireless encoder **60b** is connected to the right hand crank assembly **58b** via a right crank/encoder interface **59b**. That is, in a system according to one or more embodiments of the present disclosure, two crank assemblies **58a**, **58b** are used and two wireless encoders **60a**, **60b** are provided, which has several advantages

for the system including redundant measurement, the ability to detect a deviation (e.g., a crank assembly becoming damaged or inoperable), and the ability to safely apply a greater torque with which to open and close the IBOP valve **53**. As further shown in FIGS. **13A** and **13B**, the wireless encoders **60a**, **60b** are encased in left and right guards or covers **62a**, **62b**, as previously described. As the rotating portion **54b** of the two-piece actuator sleeve assembly **54** rotates and moves axially along the longitudinal axis of the IBOP valve **53**, as previously described, the left and right guards or covers **62a**, **62b** also move axially and rotate, according to one or more embodiments of the present disclosure.

(53) Referring now to FIGS. **15A** and **15B**, front and perspective views of an encoder assembly **75** encased within a protective guard or cover **62** are shown, according to one or more embodiments of the present disclosure. As shown in FIGS. **15A** and **15B**, the encoder assembly **75** includes a wireless encoder **60** coupled with an anti-rotation bracket **74**. As also shown in FIGS. **15A** and **15B**, the wireless encoder **60** includes a shaft **78**, which is configured to rotate, and a housing **80**, which is relatively fixed. FIG. **15C** shows a perspective view of the encoder assembly **75** encased within a transparent guard or cover **62**, which shows more details of the wireless encoder **60** coupled to the anti-rotation bracket **74**. Further, FIG. **15D** shows a perspective view of the anti-rotation bracket **74** alone. As shown in FIGS. **15A** to **15D**, the anti-rotation bracket **74** includes a plurality of rollers **76**. Referring back to FIGS. **13A** and **13B**, the plurality of rollers **76** of the anti-rotation bracket **74** facilitates axial movement of the protective guard or cover **62** relative to the wireless encoder **60** coupled with the anti-rotation bracket **74**.

(54) Still referring to FIGS. **13A** and **13B**, as the crank assemblies **58a**, **58b** rotate, e.g., 90° as previously described, the shaft **78** of the wireless encoders **60a**, **60b** also rotates in the same direction while the housing **80** of the wireless encoders **60a**, **60b** remains fixed. Moreover, the anti-rotation bracket **74** with the rollers **76** stay in the same relative position with the corresponding wireless encoders **60a**, **60b**. With the rotation of the shaft **78** as previously described, the wireless encoders **60a**, **60b** generate a plurality of measurements corresponding to the number of counts in each direction when the IBOP valve **53** is actuated. According to one or more embodiments of the present disclosure, a processing device **192**, such as that shown and described with respect to FIG. **1**, is communicatively connected with the wireless encoders **60a**, **60b**, receives the plurality of measurements from the wireless encoders **60a**, **60b**, and calculates an instantaneous position measurement of the IBOP valve **53** based on the plurality of measurements, as further described below. In this way, because the wireless encoders **60a**, **60b** generate the plurality of measurements corresponding to the number of counts in each direction when the IBOP valve **53** is actuated, the wireless encoders **60a**, **60b** are configured to provide an instantaneous position measurement of the IBOP valve **53**, as further described below.

(55) According to one or more embodiments of the present disclosure, the wireless encoders **60a**, **60b** may be powered by a battery. Moreover, the wireless encoders **60a**, **60b** according to one or more embodiments of the present disclosure are configured to operate in hazardous areas, as recognized by those having skill in the relevant art.

(56) Referring now to FIGS. **16A** and **16B**, zoomed in partial views of certain components of a top drive **28**, according to one or more embodiments of the present disclosure, are shown. For the sake of clarity, wireless encoders **60a**, **60b** and corresponding guards or covers **62a**, **62b** have been removed in FIGS. **16A** and **16B**, so that the left crank valve **58b**, as previously described, is more visible. Specifically, FIG. **16A** shows the top drive **28** in a retracted position with the left crank valve **58b** in a closed position. Due to this configuration, the IBOP valve **53** shown in FIG. **16A** is also closed. Further, FIG. **16B** shows the top drive **28** in an extended position with the left crank valve **58b** in an open position. Due to this configuration, the IBOP valve **53** shown in FIG. **16B** is also closed. With respect to FIGS. **16A** and **16B**, like components of the top drive **28** are labeled with like reference numerals, as previously described.

(57) Referring now to FIGS. **17A** and **17B**, zoomed in partial, angled cutaway views of certain

components of a top drive **28** according to one or more embodiments of the present disclosure are shown. Specifically, FIGS. **17A** and **17B** show the bearing **55** between the non-rotating portion **54a** and the rotating portion **54b** of the two-piece actuator sleeve assembly **54**, as previously described. Moreover, FIGS. **17A** and **17B** show the encoder assembly **75** connected to the left crank assembly **58b**, as previously described. Specifically, FIG. **17A** shows the top drive **28** in a retracted position with the left crank valve **58b** in a closed position. Due to this configuration, the IBOP valve **53** shown in FIG. **17A** is also closed. Further, FIG. **17B** shows the top drive **28** in an extended position with the left crank valve **58b** in an open position. Due to this configuration, the IBOP valve **53** shown in FIG. **17B** is also open. With respect to FIGS. **17A** and **17B**, like components of the top drive **28** are labeled with like reference numerals, as previously described.

(58) Referring now to FIGS. **18A** and **18B**, zoomed in partial views of certain components of a top drive **28** according to one or more embodiments of the present disclosure are shown. Specifically, FIGS. **18A** and **18B** show the protective guard or cover **62a**, **62b** over the crank valve **58a**, **58b** and the corresponding encoder assemblies **75**, for example. Moreover, FIG. **18A** shows the top drive **28** in a retracted position, and FIG. **18B** shows the top drive **28** in an extended position, as previously described. With respect to FIGS. **18A** and **18B**, like components of the top drive **28** are labeled with like reference numerals, as previously described.

(59) Referring now to FIGS. **19A** and **19B**, zoomed in partial views of certain components of the top drive **28** as shown in FIGS. **18A** and **18B**, with links **66** removed and the encoder guards or covers **62a**, **62b** transparent for better visibility, according to one or more embodiments of the present disclosure, are shown. Moreover, FIG. **19A** shows the top drive **28** in a retracted position, and FIG. **19B** shows the top drive **28** in an extended position as previously described. With respect to FIGS. **19A** and **19B**, like components of the top drive **28** are labeled with like reference numerals, as previously described.

(60) As previously described, a processing device **192** communicatively connected with wireless encoders **60a**, **60b** receives a plurality of measurements from the wireless encoders **60a**, **60b**, and calculates an instantaneous position measurement of the IBOP valve **53** based on the plurality of measurements, according to one or more embodiments of the present disclosure. An HMI may visually display position measurement data calculated by the processing device **192**, according to one or more embodiments of the present disclosure. More specifically, the processing device **192** may determine an operating extent of the IBOP valve **53** based on the instantaneous position measurement of the IBOP valve **53**, according to one or more embodiments of the present disclosure. Moreover, the determined operating extent of the IBOP valve **53** may be displayed on a screen of an HMI, such as in a driller's control room, and/or on a driller's chair, according to one or more embodiments of the present disclosure.

(61) Referring now to FIGS. **20A**, **20B**, **21A**, **21B**, **22A**, **22B**, **23A**, and **23B**, various HMI images representative of the IBOP valve **53** according to one or more embodiments of the present disclosure are shown. Specifically, FIG. **20A** shows an HMI image, which may be displayed on a screen, for example, of a new IBOP valve **53**, which is properly set up and adjusted, in a fully closed position. Indeed, the HMI image of FIG. **20A** indicates that the IBOP valve **53** has rotated the full 90°, via full rotation of the crank valve, as previously described, to assume the fully closed position. Moreover, because the IBOP valve **53** represented in FIG. **20A** is a new valve, the operating extent of the IBOP valve **53** is at full capacity.

(62) Further, FIG. **20B** shows an HMI image of a new IBOP valve **53**, which is properly set up and adjusted, in a fully open position. Indeed, the HMI image of FIG. **20B** indicates that the IBOP valve **53** has rotated the full 90° in the opposite direction to that shown in FIG. **20A**, via full rotation of the crank valve, as previously described, to assume the fully open position. Moreover, because the IBOP valve **53** represented in FIG. **20B** is a new valve, the operating extent of the IBOP valve **53** is at full capacity.

(63) FIG. **21A** shows an HMI image of a new IBOP valve **53**, which is properly set up and

adjusted, in a partially closed position. FIG. 21B shows an HMI image of a new IBOP valve 53, which is properly set up and adjusted, in a partially open position. Indeed, the HMI image of FIG. 21A indicates that the IBOP valve 53 has not rotated the full 90° in order to be fully closed, and the HMI image of FIG. 21B indicates that the IBOP valve 53 has not rotated the full 90° in the opposite direction in order to be fully opened. Instead, the IBOP valve 53 is indicated to have assumed a not fully closed position (FIG. 21A), and a not fully open position (FIG. 21B). According to one or more embodiments of the present disclosure, the system may generate an alarm upon a determination that the IBOP valve 53 is not fully closed or not fully opened.

(64) FIG. 22A shows an HMI image of a worn IBOP valve 53, which is properly set up and adjusted, in a partially closed position. FIG. 22B shows an HMI image of a worn IBOP valve 53, which is properly set up and adjusted, in a partially open position. Indeed, the HMI image of FIG. 22A indicates that the IBOP valve 53 has not rotated the full 90° in order to be fully closed, and the HMI image of FIG. 22B indicates that the IBOP valve 53 has not rotated the full 90° in the opposite direction in order to be fully opened. Instead, the IBOP valve 53 is indicated to have assumed a not fully closed position (FIG. 22A), and a not fully open position (FIG. 22B). Moreover, the HMI images of FIGS. 22A and 22B indicate that the operating extent of the IBOP valve 53 has reduced over time, as valve wear occurs. According to one or more embodiments of the present disclosure, the processing device 192 may determine the operating extent of the IBOP valve 53 based on the instantaneous position measurement of the IBOP valve 53. Moreover, the processing device 192 may track the instantaneous position measurement of the IBOP valve 53 over a period of time, and determine a condition of the IBOP valve 53 based on the instantaneous position measurement tracked over the period of time, according to one or more embodiments of the present disclosure.

(65) For example, FIG. 23A shows an HMI image of a worn IBOP valve 53, which is properly set up and adjusted, in a partially closed position, for which a replacement threshold has been met for replacing the IBOP valve 53. FIG. 23B shows an HMI image of a worn IBOP valve 53, which is properly set up and adjusted, in a partially open position, for which a replacement threshold has been met for replacing the IBOP valve 53. Indeed, the HMI image of FIG. 23A indicates that the IBOP valve 53 has not rotated the full 90° in order to be fully closed, and the HMI image of FIG. 23B indicates that the IBOP valve 53 has not rotated the full 90° in the opposite direction in order to be fully opened. Instead, the IBOP valve 53 is indicated to have assumed a not fully closed position (FIG. 23A), and a not fully open position (FIG. 23B). Moreover, the HMI images of FIGS. 23A and 23B indicate that the operating extent of the IBOP valve 53 has reduced over time so much that the IBOP valve 53 should be replaced. By the HMI images of FIGS. 23A and 23B showing that the replacement threshold of the IBOP valve 53 has been met, the user is alerted that failure of the IBOP valve 53 is imminent; however, the IBOP valve 53 has not degraded to a point that the valve will no longer hold pressure. Such an alert is advantageous because it affords the user time to plan for replacement of the IBOP valve 53 before the valve completely fails.

(66) Advantageously, systems and methods according to one or more embodiments of the present disclosure enable the addition of an IBOP actuator valve position window to the driller's HMI, the storage and trending of the measurement data, indications of IBOP valve 53 position based on direct position measurement, and the generation of alarms when specified conditions are not achieved (e.g., not fully opened, not fully closed, etc.).

(67) Although a few embodiments of the disclosure have been described in detail above, those of ordinary skill in the art will readily appreciate that many modifications are possible without materially departing from the teachings of this disclosure. Accordingly, such modifications are intended to be included within the scope of this disclosure as defined in the claims.

Claims

1. A system, comprising: a top drive comprising: a handling ring assembly comprising a main body; a main shaft that traverses the main body of the handling ring assembly; an internal blowout preventer valve, wherein the main shaft engages the internal blowout preventer valve; a two-piece actuator sleeve assembly comprising: a non-rotating portion; and a rotating portion; at least one hydraulic cylinder connected to the main body of the handling ring assembly, wherein the at least one hydraulic cylinder actuates the two-piece actuator sleeve assembly; a left hand crank assembly having a left wireless encoder connected thereto; and a right hand crank assembly having a right wireless encoder connected thereto, wherein the crank assemblies with the wireless encoders are attached to, and are configured to rotate with, the internal blowout preventer valve along with the rotating portion of the two-piece actuator sleeve assembly.
2. The system of claim 1, wherein the crank assemblies interface with a ball valve inside a body of the internal blowout preventer valve.
3. The system of claim 1, wherein the wireless encoders are powered by a battery.
4. The system of claim 1, wherein the wireless encoders are configured to provide an instantaneous position measurement of the internal blowout preventer valve.
5. The system of claim 1, wherein each wireless encoder is coupled with an anti-rotation bracket, and wherein each wireless encoder and corresponding anti-rotation bracket is encased in a protective guard.
6. The system of claim 5, wherein the anti-rotation bracket comprises a plurality of rollers, and wherein the plurality of rollers facilitates axial movement of the protective guard relative to the wireless encoder coupled with the anti-rotation bracket.
7. The system of claim 1, wherein the left wireless encoder is connected to the left hand crank assembly via a left crank/encoder interface, and wherein the right wireless encoder is connected to the right hand crank assembly via a right crank/encoder interface.
8. The system of claim 1, further comprising a bearing disposed between the non-rotating portion and the rotating portion of the two-piece actuator sleeve assembly.
9. The system of claim 1, wherein the internal blowout preventer valve comprises: a body; a box connection end; and a pin end, and wherein the main shaft engages the box connection end of the internal blowout preventer valve.
10. A method comprising: rotating the main shaft of the top drive of claim 1, thereby causing the internal blowout preventer valve and the rotating portion of the two-piece actuator sleeve assembly to rotate; applying pressure to the at least one hydraulic cylinder to actuate the two-piece actuator sleeve assembly, wherein actuating the two-piece actuator sleeve assembly comprises: moving the non-rotating portion axially along a longitudinal axis of the internal blowout preventer valve; and moving the rotating portion axially along the longitudinal axis of the internal blowout preventer valve, along with the non-rotation portion, as the rotating portion continues to rotate; rotating the crank assemblies in a first direction to shift a position of the internal blowout preventer valve, wherein, during the rotating step, a shaft of the wireless encoders corresponding to the crank assemblies rotates in the first direction; generating a plurality of measurements as the shaft of the wireless encoders rotates in the first direction; receiving the plurality of measurements by a processing device; calculating an instantaneous position measurement of the internal blowout preventer valve based on the plurality of measurements; and determining an operating extent of the internal blowout preventer valve based on the instantaneous position measurement of the internal blowout preventer valve.
11. The method of claim 10, further comprising: tracking the instantaneous position measurement of the internal blowout preventer valve over a period of time; and determining a condition of the internal blowout preventer valve based on the instantaneous position measurement tracked over the period of time.
12. The method of claim 10, wherein determining the operating extent of the internal blowout

preventer valve comprises determining whether the internal blowout preventer valve is fully opened or fully closed.

13. The method of claim 12, further comprising: generating an alarm upon a determination that the internal blowout preventer valve is not fully opened or not fully closed.

14. The method of claim 12, further comprising: indicating that the internal blowout preventer valve should be replaced based on the determined operating extent of the internal blowout preventer valve on a screen of a human machine interface.

15. The method of claim 10, further comprising: displaying the determined operating extent of the internal blowout preventer valve on a screen of a human machine interface.
